

E-Voting System Report

Design and Implementation of a Digital Voting system

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

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ECE 2.2

OBJECT ORIENTED PROGRAMMING PROJECT

1. Introduction

In the contemporary era of digital transformation, traditional electoral processes face significant challenges regarding speed, accuracy, and transparency. This project presents a C++ based Digital Voting System designed to automate the elective process for multi-tier government positions, specifically targeting MCA, MP, and Governor seats.

The system serves as a technological solution to mitigate human error in tallying, prevent electoral fraud through 'one-man-one-vote' logic enforcement, and provide instantaneous statistical outcomes through algorithmic sorting. By leveraging Object-Oriented Programming (OOP), the system ensures that voters and candidates are handled as robust data objects with strictly defined attributes.

2. System Requirements

2.1 Functional Requirements

- Functional requirements define the core actions and behaviors of the system:
 - Admin Authorization: A secure gatekeeping mechanism requiring a password (admin2026) to unlock system management features.
 - Voter Registration & Management: An interface for administrators to create unique voter records containing identity documentation and polling station metadata.
 - Identity Verification Protocol: A search-and-match algorithm that validates voter IDs against the internal database before granting access to the ballot.
 - Electoral Integrity Logic: A state-tracking mechanism that utilizes boolean flags to ensure a voter cannot re-enter the voting loop once their ballot is submitted.
 - Tiered Ballot Casting: Sequential voting cycles for MCA, Member of Parliament (MP), and Governor positions within a single session.
 - Algorithmic Tallying: Use of the Standard Template Library (STL) sort algorithm to provide real-time winner determination based on vote counts.

2.2 Non-Functional Requirements

- Security & Access Control: Restricted access to sensitive data (results and registration) ensuring system integrity.
- Data Robustness: Implementation of input buffer clearing (cin.ignore) to handle non-standard string inputs without system crashes.
- Usability: A clean, menu-driven Command Line Interface (CLI) optimized for administrative and voter clarity.
- Scalability: An architecture capable of handling expanding candidate lists using dynamic vector containers.

3. System Architecture & Design

The system follows a Modular Monolithic Architecture, organized into three distinct logical layers:

3.1 Data Layer (Classes)

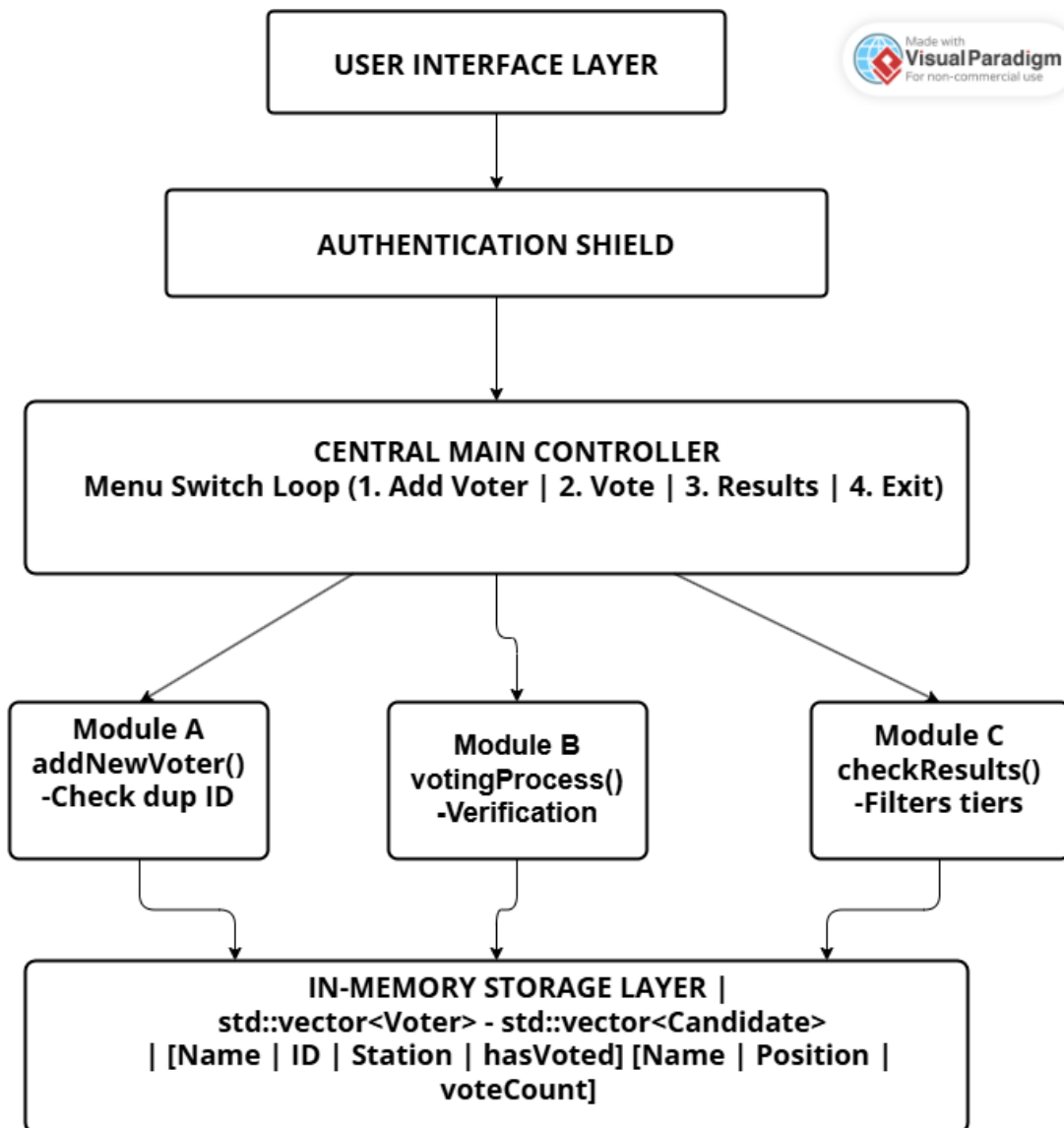
The data design utilizes Class Encapsulation and Composition. The 'Candidate' class contains an instance of the 'Party' struct, ensuring that political affiliation is intrinsically linked to the candidate data.

Class	Key Attributes	Role in System
Voter	Name, ID, Station, hasVoted	Identity tracking and eligibility enforcement.
Candidate	Name, Position, voteCount	Record of popularity and elective tier.
Party	Name, Symbol	Political context and branding for candidates.

3.2 Logic Flow

The voting logic employs a pointer-based verification system. Upon entering an ID, the system iterates through the voter vector. If a match is found, a pointer to that specific object is passed to the voting function. This allows direct modification of the 'hasVoted' status in memory, ensuring O(1) update efficiency once the voter is identified.

3.3 System Architecture Diagram



4. Output Terminal

4.1. Terminal Main menu(New voter registration)

```
=====
      DIGITAL VOTING SYSTEM
=====
1. Add New Voter
2. Voting Process
3. Check Results
4. Exit
Choice: 2

[=====

--- Voting Process ---

[=====
Enter your ID to verify: 112169998
Verification Successful. Welcome, Boniface Kimaiyo Kipchirchir.

Select candidate for MCA:
0. Jane Wanjiku (Orange Democratic Party)
1. John Omari (Universal Democratic Association)
6. Nancy Chepchumba (Wiper)
7. Rigathi Gachagua (Democrats Church Party)
12. Micheal Flexx (Collective Advisory Party)
Enter index number: 0
Vote cast for MCA.

[=====

[ADMIN ACCESS REQUIRED]
Enter Admin Password:
[=====
admin2026

=====
      DIGITAL VOTING SYSTEM
=====
1. Add New Voter
2. Voting Process
3. Check Results
4. Exit
Choice: 1

[=====

--- Register New Voter ---

[=====
Enter Full Name: Boniface Kimaiyo Kipchirchir
Enter ID Number: 112169998
Enter Polling Station: Nakuru
Voter successfully added to the system.
```

4.2. Voting Process (Voter 1 and Voter 2)

```
Select candidate for MP:
2. Alice Mutua (Orange Democratic Party)
3. Bob Kariuki (Universal Democratic Association)
8. Peter Stanley (Wiper)
9. Lamar Johnson (Democrats Church Party)
13. Chris Odinga (Collective Advisory Party)
Enter index number: 2
Vote cast for MP.

Select candidate for Governor:
4. John Mbadi (Orange Democratic Party)
5. Evans Kidero (Universal Democratic Association)
10. Esther Pasaris (Wiper)
11. Jerry Can (Democrats Church Party)
14. Mellisa Ochundo (Collective Advisory Party)
Enter index number: 4
Vote cast for Governor.

Thank you for voting! Your data has been stored.

=====
      DIGITAL VOTING SYSTEM
=====
1. Add New Voter
2. Voting Process
3. Check Results
4. Exit
Choice: 1

[=====

--- Register New Voter ---

[=====
Enter Full Name: Nick Obade
Enter ID Number: 12345678
Enter Polling Station: Naivasha
Voter successfully added to the system.

=====
      DIGITAL VOTING SYSTEM
=====
1. Add New Voter
2. Voting Process
3. Check Results
4. Exit
Choice: 2
```

Select candidate for MCA:

- 0. Jane Wanjiku (Orange Democratic Party)
- 1. John Omari (Universal Democratic Association)
- 6. Nancy Chepchumba (Wiper)
- 7. Rigathi Gachagua (Democrats Church Party)
- 12. Micheal Flexx (Collective Advisory Party)

Enter index number: 0

Vote cast for MCA.

Select candidate for MP:

- 2. Alice Mutua (Orange Democratic Party)
- 3. Bob Kariuki (Universal Democratic Association)
- 8. Peter Stanley (Wiper)
- 9. Lamar Johnson (Democrats Church Party)
- 13. Chris Odinga (Collective Advisory Party)

Enter index number: 2

Vote cast for MP.

Select candidate for Governor:

- 4. John Mbadi (Orange Democratic Party)
- 5. Evans Kidero (Universal Democratic Association)
- 10. Esther Pasaris (Wiper)
- 11. Jerry Can (Democrats Church Party)
- 14. Mellisa Ochundo (Collective Advisory Party)

Enter index number: 4

Vote cast for Governor.

Thank you for voting! Your data has been stored.

[=====

--- Voting Process ---

[=====

Enter your ID to verify: 12345678

Verification Successful. Welcome, Nick Obade.

Select candidate for MCA:

- 0. Jane Wanjiku (Orange Democratic Party)
- 1. John Omari (Universal Democratic Association)
- 6. Nancy Chepchumba (Wiper)
- 7. Rigathi Gachagua (Democrats Church Party)
- 12. Micheal Flexx (Collective Advisory Party)

Enter index number: 0

Vote cast for MCA.

Select candidate for MP:

- 2. Alice Mutua (Orange Democratic Party)
- 3. Bob Kariuki (Universal Democratic Association)
- 8. Peter Stanley (Wiper)
- 9. Lamar Johnson (Democrats Church Party)
- 13. Chris Odinga (Collective Advisory Party)

Enter index number: 2

Vote cast for MP.

4.3. Voting Results

--- Final Results & Winners ---

POSITION: MCA

Jane Wanjiku | Orange Democratic Party | Votes: 2
John Omari | Universal Democratic Association | Votes: 0
Nancy Chepchumba | Wiper | Votes: 0
Rigathi Gachagua | Democrats Church Party | Votes: 0
Micheal Flexx | Collective Advisory Party | Votes: 0
WINNER: Jane Wanjiku with 2 votes.

POSITION: MP

Alice Mutua | Orange Democratic Party | Votes: 2
Bob Kariuki | Universal Democratic Association | Votes: 0
Peter Stanley | Wiper | Votes: 0
Lamar Johnson | Democrats Church Party | Votes: 0
Chris Odinga | Collective Advisory Party | Votes: 0
WINNER: Alice Mutua with 2 votes.

POSITION: Governor

John Mbadi | Orange Democratic Party | Votes: 2
Evans Kidero | Universal Democratic Association | Votes: 0
Esther Pasaris | Wiper | Votes: 0
Jerry Can | Democrats Church Party | Votes: 0
Mellisa Ochundo | Collective Advisory Party | Votes: 0
WINNER: John Mbadi with 2 votes.